

hallucination detection 7 20 theory

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July 2026

1 Problem Setup: Key-Selection Hallucination

We study hallucinations that arise not from missing knowledge but from *key selection*: the prompt contains sufficient evidence for the correct answer, yet the model conditions its answer on a statistically salient but constraint-irrelevant span—a *shortcut key*—rather than on the span that actually determines the answer—a *constraint key*. Our goal is to (i) detect, for a given model and prompt, whether the produced answer was driven by a shortcut key, and (ii) verify causally that the identified key is responsible, first at the text level and then at the representation level.

1.1 Answers, margins, and the hallucination event

Let z denote a prompt consisting of a knowledge passage and a question. For an answer sequence y we measure the model’s support for y by the length-normalized sequence log-likelihood

$$S(y \mid z) = \frac{1}{|y|} \sum_{t=1}^{|y|} \log P_{\theta}(y_t \mid z, y_{<t}). \quad (1)$$

Two observation regimes cover our experiments. In the *paired* regime (e.g., HaluEval), each item provides a correct answer y^* and a target hallucinated answer y^h , and we define the *answer margin*

$$M(z) = S(y^* \mid z) - S(y^h \mid z), \quad H = \mathbb{K}[M(z) < 0]. \quad (2)$$

In the *open-generation* regime no paired y^h is available; the observable quantities are the generated answer $\hat{y}$, its support $S(\hat{y} \mid z)$, and a reference-based correctness label, with $H = \mathbb{K}[\hat{y} \text{ incorrect}]$. The open regime is a reference-mediated surrogate for the paired one: when $\hat{y}$ is incorrect, $S(\hat{y} \mid z)$ plays the role of $S(y^h \mid z)$, and when correct, that of $S(y^* \mid z)$. Binary-choice tasks are the special case $|y| = 1$, so a single formalism covers all our settings. Unless stated otherwise we use the length-normalized margin; the unnormalized total margin is reported in the appendix as a robustness check.

1.2 A latent key-path model and what it does (not) imply

We posit a latent path variable $\Pi \in \{F, S\}$ indicating whether the answer distribution is produced by a *faithful* path that conditions on the constraint key, or by a *shortcut* path that conditions on a shortcut key. Writing $\pi(z) = P(\Pi = S \mid z)$,

$$P_\theta(y \mid z) = (1 - \pi) P_F(y \mid z) + \pi P_S(y \mid z). \quad (3)$$

Let $a_F = P_F(y^*)$, $a_S = P_S(y^*)$, $b_F = P_F(y^h)$, and $b_S = P_S(y^h)$. We require only a *separation* condition, which is definitional rather than empirical: a putative shortcut path whose preference over (y^*, y^h) coincides with the faithful path is behaviorally unidentifiable and outside our scope.

Assumption 1 (Path separation). $a_F > a_S$ and $b_S > b_F$.

Proposition 1 (Margin monotonicity in the shortcut weight). *Under Assumption 1, the margin $M(\pi) = \log \frac{(1-\pi)a_F + \pi a_S}{(1-\pi)b_F + \pi b_S}$ is strictly decreasing in π :*

$$\frac{dM}{d\pi} = \frac{a_S - a_F}{(1 - \pi)a_F + \pi a_S} - \frac{b_S - b_F}{(1 - \pi)b_F + \pi b_S} < 0.$$

Remark 1 (Only monotonicity is implied). *The second derivative of $M(\pi)$ has indeterminate sign and may change sign on $(0, 1)$. The latent-path model therefore licenses monotonicity of the margin in the shortcut weight, but neither linearity, convexity, nor any polynomial form. This observation determines the inductive bias of our detector in Section 2: we constrain signs and monotone directions, and treat the functional form as a model-selection question.*

1.3 Spans, interventions, and operational key roles

The prompt is segmented into atomic spans $z = (s_1, \dots, s_m)$, each a candidate key. For an intervention operator $a \in \{\text{DELETE}, \text{MASK}, \text{NEUTRALIZE}\}$, let $T_i^{(a)}$ apply operator a to span s_i . Rather than deriving span importance from an assumed decomposition of internal representations, we *define* each span’s contribution as an interventional contrast—an estimand measured directly from model behavior.

Definition 1 (Interventional span contribution). *In the paired regime, the signed contribution of span s_i under operator a is $\gamma_i^{(a)} = M(z) - M(T_i^{(a)}z)$; in the open regime it is the support contribution to the generated answer, $\varphi_i^{(a)} = S(\hat{y} \mid z) - S(\hat{y} \mid T_i^{(a)}z)$. We aggregate robustly across operators, $\hat{\gamma}_i = \text{median}_a \gamma_i^{(a)}$ (analogously for $\hat{\varphi}_i$).*

A negative γ_i means that removing s_i *raises* the correct margin: the span was pushing the answer toward y^h . Roles are defined by this contrast, together with regeneration flips, which discretize large contributions.

Definition 2 (Operational roles). *Fix a minimal detectable effect $\delta > 0$. Span s_i is a shortcut ($R_i = S$) if $\hat{\gamma}_i < -\delta$ (paired), or if $\hat{\varphi}_i > \delta$ with $\hat{y}$ incorrect, or if intervention flips regeneration from incorrect to correct (open); a constraint ($R_i = C$) in the sign-symmetric cases; and irrelevant ($R_i = I$) if $|\hat{\gamma}_i| \leq \delta$ with stable regeneration.*

Two consequences of Definition 2 organize the rest of the paper. First, roles are properties of the *model’s reliance on the current prompt*, not of surface semantics: a span that “reads like” a shortcut but elicits no interventional response is irrelevant, so correctly answered items may legitimately contain cue-present-but-unused shortcut spans. Second, since interventions can have side effects (broken syntax, dangling coreference), we do not assume intervention fidelity; instead we impose a *testable* validity condition: a span enters a high-confidence role only if $\text{sign}(\gamma_i^{(a)})$ agrees across the three operators, whose side-effect profiles differ (deletion perturbs structure; masking and neutralization preserve it). Spans with inconsistent directions are marked ambiguous and excluded from role supervision.

Scope. Our theory targets knowledge-sufficient, key-selection-driven hallucination: the prompt supports y^* , a shortcut-supported y^h exists, and the error mechanism is misallocated reliance rather than missing facts. Errors caused purely by absent knowledge are outside this scope.

2 Detecting Shortcut Reliance

Interventions identify the *direction* of a span’s effect but are expensive: each span requires several prompt edits, regenerations, and scoring passes. Attention enters our pipeline in two logically distinct, non-causal roles—as an *allocation* signal deciding where to spend the intervention budget, and as a *discriminative* feature complementing behavioral signatures at test time. Neither role carries a causal claim; all causal directions come from Definition 1.

2.1 Attention as intervention allocation

Let A_i denote an answer-to-span attention score (within-item standardized mass, density, and peak over mid-to-late layers). We select the top- k spans by A_i and intervene only on those. The allocation is justified by a single weak assumption: attention need not be *calibrated* to any latent usage strength; it need only be *rank-informative* for whether an intervention will produce a detectable effect.

Assumption 2 (Rank informativeness). *For the minimal detectable effect δ of Definition 2, the map $a \mapsto P(|\gamma_i| > \delta \mid A_i = a)$ is non-decreasing.*

Proposition 2 (Optimality of top- k allocation). *Under Assumption 2, among all candidate sets C with $|C| = k$, the set $C_k^{\text{attn}} = \text{TopK}_i A_i$ maximizes the*

expected number of recovered non-null contributions, $J(C) = \mathbb{E}[\#\{i \in C : |\gamma_i| > \delta\}]$, and simultaneously minimizes the expected number of large-contribution spans left unintervened.

Proof. Write $\psi(a) = P(|\gamma_i| > \delta \mid A_i = a)$, so $J(C) = \sum_{i \in C} \psi(A_i)$ with ψ non-decreasing. If C contains i while some $j \notin C$ has $A_j > A_i$, exchanging i for j does not decrease J . Iterating yields C_k^{attn} ; the residual claim follows since the excluded mass is $\sum_{i \notin C} \psi(A_i)$. $\square$

Assumption 2 is directly testable, and we test it: binning spans by A_i and checking that $\mathbb{E}[|\hat{\gamma}_i|]$ increases across bins, reporting $\text{Spearman}(A_i, |\hat{\gamma}_i|)$, and comparing top- k against random- k selection on total recovered effect mass. The framework is modular in this assumption: if it failed, screening would degrade to random sampling, costing sample efficiency and coverage (Section 3.1) but leaving Definitions 1–2 and the detector below intact.

2.2 Reference-free role inference

At training time, Definition 2 is instantiated with references to produce pseudo-roles with reliability weights. At test time no reference is available; a classifier maps the behavioral signature B_i (per-operator contrasts, flips, similarity, and entropy changes) and attention features A_i^{feat} (layerwise mass/density/peak and their intervention differentials) to a role posterior $q_i(r) = P(R_i = r \mid B_i, A_i^{\text{feat}})$, $r \in \{\text{S}, \text{C}, \text{I}\}$. The detector thus estimates the latent reliance variable, not the correctness label; references enter only through training supervision.

Including attention as a feature requires no assumption. By the chain rule,

$$I(R_i; B_i, A_i^{\text{feat}}) = I(R_i; B_i) + I(R_i; A_i^{\text{feat}} \mid B_i) \geq I(R_i; B_i), \quad (4)$$

with strict inequality iff the conditional term is positive—an *empirical* quantity that our feature ablation estimates directly. The conditional term is plausibly positive for three reasons: behavioral contrasts are noisy for small-effect spans, whereas attention provides pre-intervention evidence of use; redundant spans mask each other behaviorally (deleting one is compensated by the other) while both remain visible to attention; and behavioral contrasts exist only for the k selected spans, whereas attention scores exist for all. Empirically, adding attention to behavior raises item-level test AUROC from 0.824 to 0.883 on ScientistQA, consistent with a strictly positive conditional term; the spectral variant (0.870) confirms the gain is not specific to raw attention, and we report it as an ablation.

2.3 Evidence aggregation and the monotone detector

Definition 3 (Usage and item-level evidence). *Let $u_i \in [0, 1]$ be a bounded plug-in for $|\hat{\gamma}_i|$ (a saturating transform of per-operator contrasts plus flip and*

contradiction terms). Define

$$E_s = \frac{\sum_i u_i q_i(S)}{\sum_i u_i}, \quad E_c = \frac{\sum_i u_i q_i(C)}{\sum_i u_i}, \quad M_s = \max_i u_i q_i(S), \quad K_s = \text{topk-mean}_i u_i q_i(S). \quad (5)$$

Normalizing by total usage (rather than span count) makes E_s the *fraction of active reliance flowing to shortcuts*, comparable across items with different span counts and overall perturbation sensitivity. The peak statistics M_s and K_s matter because key-selection failures are often driven by a single dominant key that mean evidence dilutes; all four statistics are symmetric, coordinatewise-monotone aggregations of $\{u_i q_i(S)\}$, so they preserve the monotone structure below.

The link from evidence to hallucination requires only a sign condition on the map from evidence to the shortcut weight.

Assumption 3 (Sign-constrained evidence link). $\text{logit } \pi = g(E_s, E_c)$ for some differentiable g with $\partial g / \partial E_s > 0$ and $\partial g / \partial E_c < 0$.

Proposition 3 (Monotone detector structure). *Under Assumptions 1 and 3 and Proposition 1, $\partial M / \partial E_s < 0$ and $\partial M / \partial E_c > 0$; consequently, for $H = \mathbb{K}[M + \xi < 0]$ with evidence-independent noise ξ ,*

$$\frac{\partial P(H = 1)}{\partial E_s} > 0, \quad \frac{\partial P(H = 1)}{\partial E_c} < 0.$$

Proposition 3 is the *entire* constraint the theory places on the detector: sign monotonicity, and nothing about functional form (Remark 1). We therefore fit within the sign-constrained monotone family and use as our primary model its minimal parametric member,

$$\text{logit } \hat{P}(H = 1) = b + \beta_1 E_s - \beta_2 E_c + \beta_3 M_s + \beta_4 K_s, \quad \beta_j \geq 0, \quad (6)$$

with the nonnegativity enforced by a softplus reparameterization. Linearity is a good approximation exactly when the two paths are nearly separable ($a_F, b_S \approx 1$ and $a_S, b_F \approx 0$), in which case $M(\pi) \approx -\text{logit } \pi$; in general, richer monotone members (e.g., a monotone GAM $b + f_s(E_s) + f_c(E_c)$ with $f'_s \geq 0$, $f'_c \leq 0$) are compared under nested cross-validation on held-out log loss and calibration, with the sign constraints held fixed as the inductive bias rather than selected over.

Fitted on 500 constrained-QA items, the learned coefficients respect the predicted signs without being forced to particular magnitudes ($b = -0.331$, $\beta_1 = 4.32$, $\beta_2 = 2.48$), and shortcut evidence alone separates hallucinated from correct items (mean E_s : 0.247 vs. 0.106; bootstrap 95% CI of the difference [0.089, 0.196]; single-feature AUROC 0.804)—the qualitative pattern $E_s(\text{hallucinated}) > E_s(\text{correct})$ predicted by Proposition 3. Because q_i and u_i are exposed per span, the detector is interpretable by construction: each item’s score decomposes into signed span contributions, naming the span that carries the largest shortcut evidence—which is precisely the handle the mitigation stage requires.

3 Mitigating and Causally Validating Shortcut Reliance

The detector of Section 2 makes a falsifiable claim: *this* span is the shortcut key the model currently relies on. We test the claim with two interventions derived from the same latent-path model of Section 1.2—a text-level manipulation of the key’s surface prior (KeyShift) and a representation-level state transplant—and use the first as a detector-gated mitigation.

3.1 KeyShift: manipulating the surface prior of the key

Semantically equivalent realizations of the same key need not carry the same pretraining-induced pull toward the hallucinated answer. We operationalize this *surface prior* behaviorally, via an isolated-span probe that asks the model to answer using only the candidate realization e :

$$\hat{\rho}(e) = S(y^h \mid e, q) - S(y^* \mid e, q), \quad (7)$$

the shortcut margin the realization induces *on its own*. Paraphrase candidates for the detector-identified shortcut span are generated, filtered by a validator (entity, number, polarity, and edit-ratio checks plus an LLM review), and ranked by $\hat{\rho}$. *Prior-directed paraphrasing* (PDP) selects the valid realization minimizing $\hat{\rho}$; *prior-high* selects the maximizer; a *common control* matches the original’s prior to isolate the prior channel from generic rewording effects. We do not assume paraphrases are perfectly meaning-preserving:

Assumption 4 (Bounded semantic drift). *Within the validated equivalence class, replacements preserve (y^*, y^h) and change the path-conditional answer distributions by at most ε_{sem} , an error estimated by validator pass rates and human audit rather than assumed to vanish.*

Corollary 1 (KeyShift predictions). *Suppose ρ enters the shortcut weight through a monotone link (as E_s does in Assumption 3). Then under Proposition 1 and Assumption 4: (i) within an item, $\partial M / \partial \rho < 0$, so the within-item fixed-effect slope of the full-prompt margin on $\hat{\rho}$ is negative; (ii) $M(\text{prior-low}) > M(\text{prior-high})$, and the PDP gain over the common control isolates the prior channel; (iii) a context-link edit that lowers the contextual support χ for the shortcut path combines with PDP so that $M_{\text{joint}} \geq \max(M_{\text{PDP}}, M_{\text{context}})$. No super-additivity is predicted, since it would depend on the unknown curvature of the link.*

Prediction (i) is primary: it is continuous, uses every valid paraphrase, and does not pass through an accuracy threshold. Empirically the within-item fixed-effect slope is -0.192 with bootstrap 95% CI $[-0.391, -0.037]$, excluding zero in the predicted direction; prior-low variants outperform prior-high and common-control variants, and the joint edit is the strongest condition, as (ii)–(iii) predict.

Detector-gated mitigation and coverage. As a mitigation, PDP is applied only when the pipeline event $I = L \wedge C \wedge D$ occurs: the shortcut is localized (L), a valid counterfactual is constructible (C), and the detector fires (D). The population effect then obeys the exact decomposition

$$\mathbb{E}[\Delta M_{\text{global}}] = P(I) \cdot \mathbb{E}[\Delta M \mid I = 1], \quad (8)$$

so global gains factor into *coverage* times *conditional efficacy*. Our current instantiation is high-precision, low-coverage: on the gated subset, three of four intervened items flip from incorrect to correct with zero correct-to-incorrect flips, yet $P(I) = 3.2\%$ caps the full-split accuracy gain at 2.4 points. Equation (8) attributes the gap to coverage—itsself decomposable into top- k screening residue (Proposition 2), counterfactual constructibility, and detector recall—rather than to any failure of conditional efficacy, and identifies coverage as the binding constraint for future work.

3.2 Representation-level validation

If the detector’s attention-based usage claim is right—the identified span’s information actually enters the answer computation—then the original→PDP margin shift should be partially carried by internal states. Let r^O and r^P denote pre-output-projection attention-head states on the original and PDP prompts, $\Delta r = r^P - r^O$, and $M = f(r)$ locally smooth. A first-order expansion gives, for head j ,

$$f(r_{-j}^O, r_j^P) - f(r^O) = \langle \nabla_{r_j} f(r^O), \Delta r_j \rangle + O(\|\Delta r_j\|^2), \quad (9)$$

so the gradient×activation-difference attribution is the first-order approximation of a single-head patch effect and serves as a principled pre-screen—unlike raw attention differences, which indicate reading but not effect. For a screened head subset $\mathcal{S}$ with projector $P_{\mathcal{S}}$, our evidence standard is *bidirectional consistency* against layer-matched random-head baselines: the forward patch $F_{\mathcal{S}} = f(r^O + P_{\mathcal{S}}\Delta r) - f(r^O)$ must reproduce part of the improvement, and the reverse rescue $R_{\mathcal{S}} = f(r^P) - f(r^P - P_{\mathcal{S}}\Delta r)$ must undo part of it. Transplantation is preferred over destructive ablation, whose effects are confounded by redundancy-driven compensation; consistently, in our cross-fitted evaluation the target-minus-random CIs for both $F_{\mathcal{S}}$ and $R_{\mathcal{S}}$ exclude zero while the ablation CI does not, and the symmetric ratio $(F_{\mathcal{S}} + R_{\mathcal{S}})/(2\Delta M_{\text{PDP}})$ averages 0.29.

Remark 2 (What is and is not claimed). *The ratio above is a mediation-fraction proxy, not a formal natural indirect effect: the network is nonlinear and heads interact. Moreover, exact head identities are unstable across folds, as expected when redundant heads span a shared functional subspace—any of several subsets can transplant approximately the same direction. Our claim is therefore about a sparse, layer-localized pathway that partially mediates the prior-induced margin shift, not about a unique “hallucination head”; the remainder is presumably carried by MLPs, the residual stream, unselected heads, and earlier positions.*

Together, Corollary 1 and the bidirectional patching results close the loop opened in Section 1: the span the detector names as the shortcut key is causally implicated at the text level, and the shift induced by manipulating its surface prior is partially traceable through the model’s internal states.